

COURSE STRUCTURE & SYLLABUS

(Common to B.Tech - CSE, CSE(DS), CSE(AI&ML), CSE(AI),
CSE(AI&DS),CSE(CS),CSIT,CSE(IOT, CS & BT)

For Admitted Batch of 2025

II YEAR I SEMESTER

Subject Code	Course Category	Course Title	L	T	P	C
25PSS301	BS	Probability and Statistics	3	0	0	3
25DAA301	PC	Design and Analysis of Algorithms	3	1	0	4
25ACS301	HS	Advanced Communication Skills	1	0	2	2
25DES301	ES	Digital Electronics	3	0	0	3
25BWT301	PC	Back End Web Technologies	3	1	0	4
25LAS301	ES	Logical Reasoning and Analytical Skills	1	0	2	2
25DAA301L	PC	Design and Analysis of Algorithms Lab	0	0	2	1
25BWD301L	PC	Backend Web Development Lab	0	0	2	1
25COI301	HS	Constitution of India	0	0	1	0
Total			14	2	9	20

25PSS301: PROBABILITY AND STATISTICS

B.Tech. II Year I Semester

L	T	P	C
3	0	0	3

COURSE OUTCOMES

CO1: Ability to summarize and visualize datasets using descriptive statistics.

CO2: Ability to analyze relationships via correlation and regression techniques and build predictive models.

CO3: Ability to apply probability rules, conditional probability, Bayes' theorem and counting principles to uncertain events.

CO4: Ability to understand and apply discrete and continuous probability distributions using PMF, PDF and CDF to solve real-world problems.

CO5: Ability to perform statistical inference using sampling techniques, confidence intervals and hypothesis testing.

MODULE I: DESCRIPTIVE STATISTICS

Introduction to Statistics, Importance, Data Visualization Using Frequency Tables, Measures of Central Tendency (Mean, Median, Mode), Measures of Dispersion (Range, IQR, Variance, Standard Deviation).

MODULE II: CORRELATION AND REGRESSION

Correlation, Covariance, Types of Correlation, Coefficient of Correlation, Simple Linear Regression, Multiple Linear Regression, Nonlinear Regression.

MODULE III: FUNDAMENTALS OF PROBABILITY

Sample Space, Events, Counting Sample Points, Probability of an Event, Additive Rules, Conditional Probability, Independence, and the Product Rule, Law of Total Probability, Bayes' Theorem. Random Variables (Discrete and Continuous)

MODULE IV: PROBABILITY DISTRIBUTIONS

Bernoulli Distribution, Binomial Distribution, Poisson Distribution, Uniform Distribution with their properties and applications, Probability Mass Function (PMF), Probability Density Function (PDF)

MODULE V: INFERENCE STATISTICS

Population and Sample, Parameter and Statistics, Sampling Techniques (Probabilistic and Non-probabilistic), Central Limit Theorem, Confidence Intervals, Hypothesis Testing (Null and Alternative Hypotheses, Type I and Type II Errors, Significance Level).

TEXT BOOKS:

1. Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, Keying Ye, Probability & Statistics for Engineers & Scientists, 9th Ed. Pearson Publishers.
2. William Mendenhall, Robert J. Beaver & Barbara M. Beaver, Introduction to Probability and Statistics

REFERENCE BOOKS:

1. T. T. Soong, Fundamentals of Probability and Statistics for Engineers, John Wiley & Sons, Ltd, 2004.
2. Sheldon M Ross, Probability and statistics for Engineers and scientists, academic press.

25DAA301: DESIGN AND ANALYSIS OF ALGORITHMS

B.Tech. II Year I Semester

L	T	P	C
3	1	0	4

COURSE OUTCOMES

CO1: Able to Apply space and time complexity analysis using asymptotic notations.

CO2: Able to Design divide-and-conquer algorithms and critically assess their runtime and space trade-offs.

CO3: Able to Device backtracking and dynamic programming solutions.

CO4: Able to Apply greedy methods and graph traversal algorithms

CO5: Able to Analyze and Design branch-and-bound algorithms for NP-hard problems

MODULE - I

Introduction: Algorithm, Performance Analysis-Space complexity, Time complexity

Divide and conquer: General method, applications-Binary search, Quick sort, Merge sort, Strassen's matrix multiplication.

MODULE - II

Disjoint Sets: Disjoint set operations, union and find algorithms, Priority Queue- Heaps, Heap sort

Backtracking: General method, applications, n-queens problem, sum of subsets problem, graph coloring, Hamiltonian cycles.

MODULE - III

Dynamic Programming: General method, applications- Optimal binary search tree, 0/1 knapsack problem, All pairs shortest path problem, Traveling salesperson problem, Reliability design.

MODULE - IV

Greedy method: General method, applications- Job sequencing with deadlines, knapsack problem, Minimum cost spanning trees, Single source shortest path problem. Basic Traversal and Search Techniques: Techniques for Binary Trees, Techniques for Graphs, Connected components.

MODULE - V

Branch and Bound: General method, applications - Travelling salesperson problem, 0/1 knapsack problem - LC Branch and Bound solution, FIFO Branch and Bound solution.

NP-Hard and NP-Complete problems: Basic concepts, non-deterministic algorithms, NP - Hard and NP- Complete classes, Cook's theorem.

TEXT BOOKS:

1. Fundamentals of Computer Algorithms, Ellis Horowitz, Satraj Sahni, and Rajasekaran, University Press.

REFERENCE BOOKS:

1. Design and Analysis of algorithms, Aho, Ullman, and Hopcroft, Pearson education.

2. Algorithm Design: Foundations, Analysis and Internet Examples, M.T. Goodrich and R. Tamassia, John Wiley and Sons.

25ACS301: ADVANCED COMMUNICATION SKILLS

B.Tech.II Year I Semester

L	T	P	C
1	0	2	2

COURSE OUTCOMES

CO1: To develop grammatical accuracy and vocabulary control for effective communication in academic and workplace contexts.

CO2: To enhance clarity and coherence in oral and written expression through sentence variety and logical organization.

CO3: To cultivate questioning, reasoning and discussion techniques for purposeful and engaging interaction.

CO4: To refine persuasive and influential communication using appropriate vocabulary, expressions, and tone. To build confidence and professionalism in public speaking, presentations and formal exchanges.

CO5: To enable students to use correct grammatical structure especially Tenses, to communicate ideas clearly, and know the importance of Letter Writing.

MODULE I: ENGLISH FOR DISCUSSIONS

Language Skills for Effective Group Interaction

Understanding others' viewpoints in group interaction, Identifying key ideas, supporting details, and transitions, Listening for tone, clarity, and appropriateness, Grammatical accuracy and fluency in discussions & Structuring sentences for coherence and variety , Using clauses, phrases, and linking devices, Maintaining clarity, tone, and appropriateness, Reading discussion-based texts and dialogues & Identifying logical flow and coherence in arguments, Analyzing sentence structure and transitions, Organizing ideas logically for discussions: Writing coherent paragraphs using clauses and transitions, Practicing structured discussion summaries

MODULE II: QUESTIONING TECHNIQUES

Academic Questioning and Response Skills

Understanding different types of questions & Interpreting context-based and academic questions, Listening to responses for accuracy and intent, Framing clear, relevant, and purposeful questions Using modals for permission, obligation, and possibility & Responding with precision, clarity, and confidence, Reading academic texts to identify

implicit and explicit questions & Understanding question patterns and discussions, Writing well-structured academic questions: Paraphrasing and reporting responses accurately, Summarizing question-answer exchanges

MODULE III: LANGUAGE OF PERSUASION

Persuasive Communication and Argumentation

Listening to persuasive speeches and debates: Identifying arguments, counterarguments, and strategies, Structuring persuasive discourse effectively, presenting arguments and counterarguments confidently, Maintaining fluency and contextual suitability. Reading persuasive texts, editorials, and opinion pieces, Identifying credibility, tone and stylistic coherence, Writing persuasive paragraphs and essays & Using appropriate vocabulary, idioms and collocations, Developing coherent arguments with supporting evidence.

MODULE IV: PUBLIC SPEAKING SKILLS

Oral Presentation and Delivery Skills

Listening to speeches for stress, rhythm, and intonation & Observing verbal and non-verbal cues of speakers, Using stress, rhythm, and intonation effectively, Organizing and structuring presentations & Ensuring tense, voice and grammatical consistency, Reading presentation scripts and speech models .Analyzing structure, flow, and delivery techniques, Preparing presentation outlines and scripts. Writing introductions, conclusions and transitions, Editing content for clarity and impact.

MODULE V: GRAMMAR OF COMMUNICATION

Functional Grammar and Written Communication

Understanding the concept of time in spoken contexts: Identifying tense usage in formal and informal communication, Using tenses correctly in spoken communication, Practicing formal and informal language in conversations. Reading formal and informal letters. Identifying grammatical patterns and tense usage, Using tenses correctly in writing. Writing formal and informal letters: Developing effective written communication skills.

TEXT BOOKS:

1. Communicative English for Engineers and Professionals – Nitin Bhatnagar & Mamta Bhatnagar
2. Professional Communication – Meenakshi Raman & Sangeeta Sharma
3. English and Communication Skills for Students of Science and Engineering – S. P. Dhanavel

Alternative NPTEL/SWAYAM COURSE:

S.NO.	NPTEL Course Name	Instructor	Host Institute
1	Technical Communication for Engineers	Prof. Arun K. Saraf	IIT Roorkee

25DES301: DIGITAL ELECTRONICS

B.Tech. II Year I Semester

L	T	P	C
3	0	0	3

COURSE OUTCOMES

- C01:** Apply Boolean algebra and minimization techniques to simplify Boolean functions.
- C02:** Minimize combinational functions and design the combinational using logic gates.
- C03:** Design and analyze combinational Logic circuits
- C04:** Construct synchronous sequential circuits combining flip-flops and logic gates.
- C05:** Utilize memory and programmable logic devices in digital system design.

MODULE I

Number Systems: Binary, Octal, Decimal, Hexadecimal, Complements of Numbers: 1's and 2's Complement, Boolean algebra: Basic theorems and properties, Boolean Functions: Canonical and standard forms, Logic Gates: AND, OR, NOT, NAND, NOR, Exclusive-OR and Exclusive-NOR, Universal Gates.

MODULE II

Gate-Level Minimization Techniques: Karnaugh maps: 2 and 3 variables, Sum-of-products (SOP) and product-of-sums (POS) simplification, Don't care conditions, Implementation using NAND and NOR gates.

MODULE III

Combinational Logic Circuits: Design procedure, Half adder, Full Adder, Half subtractor – Full subtractor BCD adder, magnitude comparator, decoders, encoders, multiplexers and demultiplexers.

MODULE IV

Sequential Logic Circuits: Latches, Flip-flops: Clocked S-R, D, T, JK, Master-Slave JK, synchronous and asynchronous counters, Ring counter, Shift registers-Left, Right Register, applications of shift register.

MODULE V

Memory Devices: Programmable Logic Devices: Memory devices - RAM, ROM, PROM, EPROM, EEPROM, EAPROM, Programmable Logic Arrays (PLA), and Programmable Array Logic (PAL).

TEXT BOOKS:

1. M. Morris Mano, Michael D. Ciletti, Digital Design with an Introduction to the Verilog HDL, 6th Edition, Pearson Education/PHI, 2017.
2. R. P. Jain - Modern Digital Electronics, 3rd Edition, 2007- Tata McGraw-Hill

REFERENCE BOOKS:

1. Ronald J. Tocci, Neal S. Widmer, Gregory L. Moss, Digital Systems: Principles and Applications, 10th Edition, Pearson Education.
2. Charles H. Roth Jr., Larry L. Kinney, Fundamentals of Logic Design, 6th Edition, Cengage Learning.
3. John.M Yarbrough, "Digital Logic Applications and Design", Thomson Learning

25BWT301: BACKEND WEB TECHNOLOGIES

B.Tech. II Year I Semester

L	T	P	C
3	1	0	4

COURSE OUTCOMES

C01: Build and configure backend servers using Node.js and Express with proper routing and error handling.

C02: Design databases, perform CRUD operations using SQLite, and develop RESTful APIs.

C03: Implement secure authentication systems using JWT and middleware.

C04: Handle file operations and manage data efficiently using streams and buffers.

C05: Develop complete backend applications using MongoDB, Mongoose, and role- based access control.

MODULE I: NODE .JS and Express Fundamentals

Introduction to Backend Development, Node.js Basics, Common JS and ES Modules, Express.js Introduction, HTTP Server Setup, Handling HTTP Requests, Server Architecture, Development Environment Setup, NPM Package Management, Routing Basics, Request and Response Objects, Error Handling.

MODULE II: Database Integration and Restful API Development

SQLite Basics and Integration, Database Schema , CRUD Operations, Query Writing and Optimization, RESTful API Development, Query Handling, REST Principles, Request Validation, Response Formatting, Testing APIs, Debugging Techniques.

MODULE III: Authentication and Middleware

JWT Token Authentication, Authentication Strategies, User Management, Middleware Functions and Implementation, Custom Middleware Development, Security Best Practices, Error Logging, Authorization Controls.

MODULE IV: File Handling and Streams

File Handling using the fs Module, File System Operations, Working with File Streams, Data Transformation Streams, Error Handling with Streams, Stream Piping, Buffer Management, Asynchronous File Operations.

MODULE V: MONGODB and NODE JS Integration

MongoDB Installation and Setup, MongoDB Atlas Integration, Native Driver vs Mongoose, Schema Design, Defining Models, Filtering and Sorting, CRUD APIs with MongoDB, Password Hashing, JWT Authentication, Role-Based Authorization, Testing MongoDB-Backed APIs, Debugging.

TEXT BOOKS:

1. Web Development with Node and Express by Ethan Brown
2. Node.js Design Patterns by Mario Casciaro and Luciano Mammino
3. RESTful Web APIs by Leonard Richardson and Mike Amundsen

REFERENCE BOOKS:

1. MongoDB: The Definitive Guide by Kristina Chodorow

25LAS301: LOGICAL REASONING AND ANALYTICAL SKILLS

B.Tech. II Year I Semester

COURSE OUTCOMES

L	T	P	C
1	0	2	2

C01: Solve arrangement problems using structured logic

C02: Analyze data sufficiency and blood relations

C03: Apply Venn diagrams and syllogisms

C04: Solve complex puzzles and evaluate logic

C05: Interpret visual patterns and spatial problems

MODULE I

Data Arrangements: Linear and circular arrangements, seating arrangements (row and column-based), grouping and sequencing and conditional arrangements.

MODULE II

Data Sufficiency & Blood Relations: Assessing whether given statements are sufficient to answer a question, covering topics like numbers, arrangements, ages and logical reasoning. Blood Relations: Family tree-based problems, paternal and maternal relationships, coded blood relations and generation-based questions.

MODULE III

Set Theory & Basic Syllogisms: Venn Diagrams: Set-based representation, logical categorization, solving problems involving union, intersection and complements of sets. Syllogisms: Statements and conclusions, logical deduction using Venn diagrams.

MODULE IV

Validity & Logical Reasoning Puzzles: Evaluating the validity of conclusions and solving categorical syllogism problems. Puzzles: Logical reasoning-based puzzles such as floor-based, box arrangement, scheduling, classification and grid-based puzzles.

MODULE V

Shapes and Pattern Deduction: Identifying and counting geometric shapes like triangles, squares, rectangles and other patterns within a complex figure. **Visual Reasoning:** Mirror and water images, paper folding and cutting, pattern completion, figure analogy and odd-one-out identification.

TEXT BOOKS

1. A Modern Approach to Verbal & Non-Verbal Reasoning – R. S. Aggarwal
2. Analytical Reasoning – M. K. Pandey

REFERENCE BOOKS

1. How to Prepare for Logical Reasoning – Arun Sharma
2. Logical Reasoning and Data Interpretation

25DAA301L: DESIGN AND ANALYSIS OF ALGORITHMS LAB

B.Tech. II Year I Semester

L	T	P	C
0	0	2	1

LIST OF EXPERIMENTS

1. Write a python program to implement Quick sort algorithm for sorting a list of integers in ascending order
2. Write a python program to implement Merge sort algorithm for sorting a list of integers in ascending order.
3. Write a python program to implement the Depth First Search algorithm for a graph. ii) Write a python program to implement the Breadth First Search algorithm for a graph.
4. Write a python programs to implement backtracking algorithm for the N- queens problem.
5. Write a python program to implement the backtracking algorithm for the sum of subsets problem.
6. Write a python program to implement the backtracking algorithm for the Hamiltonian Circuits problem.
7. Write a python program to implement greedy algorithm for job sequencing with deadlines.
8. Write a python program to implement Dijkstra's algorithm for the Single source shortest path problem.
9. Write a python program that implements Prim's algorithm to generate minimum cost spanning tree.
10. Write a python program that implements Kruskal's algorithm to generate minimum cost spanning tree
11. Write a python program to implement Floyd's algorithm for the all pairs shortest path problem.
12. Write a python program to implement Dynamic Programming algorithm for the 0/1 Knapsack problem.

25BWD301L: BACKEND WEB DEVELOPMENT LABORATORY

B.Tech. II Year I Semester

L	T	P	C
0	0	2	1

LIST OF EXPERIMENTS

1. Student Management REST API using Node.js, Express, and SQLite with CRUD operations and validation
2. User Authentication System with registration, login, password hashing and JSON Web Token
3. Full CRUD API using MongoDB and Mongoose
4. File Upload and Management System using Express and fs module
5. Blog API with authentication, CRUD operations, and role-based authorization
6. API validation and testing project using middleware and Postman
7. Logging and error handling system using custom middleware and file logging
8. Stream-based file copier using Node.js streams and buffer handling
9. Role-based access control system using JWT and middleware

25COI301: CONSTITUTION OF INDIA

B.Tech. II Year I Semester

L	T	P	C
0	0	1	0

COURSE OUTCOMES

CO1: Discuss the growth of the demand for civil rights in India for the bulk of Indians before the Arrival of Gandhi in Indian politics.

CO2: Discuss the intellectual origins of the framework of argument that informed the Conceptualization of social reforms leading to revolution in India.

CO3: Discuss the circumstances surrounding the foundation of the Congress Socialist Party [CSP] under the leadership of Jawaharlal Nehru and the eventual failure of the proposal of direct Elections through adult suffrage in the Indian Constitution

CO4: Discuss the passage of the Hindu Code Bill of 1956.

CO5: Understand the importance of constitutional and statutory bodies in safeguarding democratic values, social justice, and equality.

MODULE - I: History of Making of the Indian Constitution- History of Drafting Committee, Philosophy of the Indian Constitution- Preamble Salient Features

MODULE II: Contours of Constitutional Rights & Duties - Fundamental Rights

- Right to Equality
- Right to Freedom
- Right against Exploitation
- Right to Freedom of Religion
- Cultural and Educational Rights
- Right to Constitutional Remedies
- Directive Principles of State Policy
- Fundamental Duties.

MODULE III: Organs of Governance: Parliament, Composition, Qualifications and Disqualifications, Powers and Functions, Executive, President, Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions.

MODULE IV: Local Administration: District's Administration head: Role and Importance, **Municipalities:** Introduction, Mayor and role of Elected representative, CEO of Municipal Corporation.

Panchayat Raj: Introduction, PRI: Zila Panchayat. Elected officials and their roles, CEO ZilaPanchayat: Position and role. Block level: Organizational Hierarchy (Different departments), **Village level:** Role of Elected and Appointed officials, Importance of grass root democracy

MODULE V: Election Commission: Role and Functioning. Chief Election Commissioner and Election Commissioners. State Election Commission: Role and Functioning. Institute and Bodies for the welfare of SC/ST/OBC and women.

TEXT BOOKS

1. The Constitution of India, 1950 (Bare Act), Government Publication.
2. Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.
3. M. P. Jain, Indian Constitution Law, 7th Edn., Lexis Nexis, 2014.
4. D.D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2015.